

CS5322 Database Security Project I

Virtual Private Database for a University Information System

Project Report

This report presents the application design, security policies, Oracle VPD implementation, query tests, evaluation and demonstration plan for Project I.

Verified environment: Oracle AI Database Free Lite 23.26.3.0.0 running in Docker on WSL2

CS5322 Project Team

1 Introduction

This project implements Oracle Virtual Private Database (VPD) for a university information system. The system stores students, professors, departments, courses, enrolments, grades and payments. The same database is shared by several categories of users, so table-level grants alone are insufficient: different users must see different rows of the same table. VPD appends a policy predicate to queries at runtime, enforcing row-level security without requiring every application query to repeat the filter.

The implementation was deployed and verified in Oracle AI Database Free Lite 23.26.3.0.0 inside Docker on WSL2. The final acceptance script reported five enabled policies and all expected row counts.

2 Application and Users

The application is a university information system. It supports students, professors, department administrators, finance officers and university administrators. Each role has a different business purpose and therefore a different data view.

Role	Business need	VPD scope
Student	View personal academic and payment records	Own student, enrolment, grade and payment rows
Professor	Manage teaching and grades	Own courses and related enrolments/grades
Department Admin	Manage departmental academic data	Rows belonging to own department
Finance	Process fees and payments	All payments; no academic grades
University Admin	Institution-wide oversight	All project rows

3 Database Design

The schema contains DEPARTMENT, APP_USER, STUDENT, PROFESSOR, COURSE, ENROLLMENT, GRADE and PAYMENT. APP_USER maps the database/application username to a role and department. Foreign keys connect courses to professors, enrolments to students and courses, grades to enrolments, and payments to students. The supplied dataset contains three departments, three students, two professors, four courses, four enrolments, four grades and three payments.

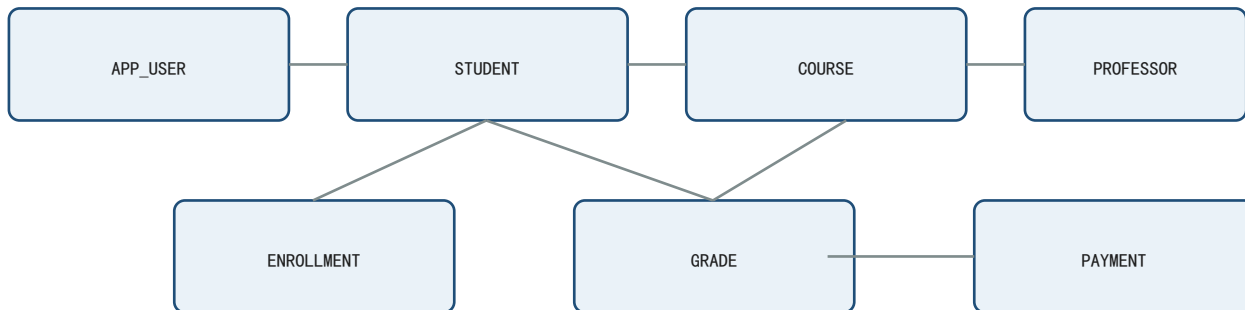


Figure 1. Core entities and relationships used by the VPD project.

The schema script also creates indexes for the main lookup columns. The unique constraint on GRADE.ENROLLMENT_ID already creates an index, so no redundant index is created.

4 Security Requirements

The requirements are deliberately nontrivial because the same tables are shared by users with different scopes.

No.	Requirement	Protected data
1	A student can read only personal records.	STUDENT, ENROLLMENT, GRADE, PAYMENT
2	A professor can read/update grades for own courses.	COURSE, ENROLLMENT, GRADE
3	A department admin manages own department data.	STUDENT, COURSE, ENROLLMENT, GRADE
4	Finance can read payments but not grades.	PAYMENT vs GRADE
5	University Admin can read all rows.	All protected tables
6	Invalid/missing identity exposes no protected rows.	All VPD tables
7	Predicates apply to reads and writes.	SELECT, INSERT, UPDATE, DELETE

5 VPD Design and Implementation

After trusted authentication, the application sets the context package to CS5322_SECURITY_CTX.SET_USER. The package sets the USER_ID, ROLE and DEPARTMENT_ID in the CS5322_SECURITY_CTX application context. Policy functions read the context and return a SQL predicate. Normal users can only activate only their own identity; the schema owner can activate any identity; the exception is used for repeatable classroom tests.

Five policy functions are registered using DBMS_RLS.ADD_POLICY: STUDENT_VPD_FN, COURSE_VPD_FN, ENROLLMENT_VPD_FN, GRADE_VPD_FN and PAYMENT_VPD_FN. A policy is enabled for SELECT, INSERT, UPDATE and DELETE. A role that should not access a table receives a policy with 1=0.

```
Example predicates:\nStudent: user_id = current_user_id\nDepartment admin: department_id = current_department_id\nProfessor grade: enrollment_id belongs to a course taught by professor\nFinance grade access: no access\nand therefore no rows
```

6 Implementation and Reproducibility

The implementation is separated into privilege-aware scripts. The schema owner creates the tables, context and package. SYSDBA creates application users and roles. The VPD script must be executed as CS5322_P1 so that the policy functions are owned by the same schema named by the policies. If tables are deleted and recreated, Oracle removes grants on the old table objects; the regrant script restores the role permissions.

```
00_create_schema.sql create schema and quota\n01_schema_and_data.sql tables, context, package and data\n02_users_roles.sql users, roles and grants\n03_regrant_object_privileges.sql grants after table reset\n04_vpd_policies.sql VPD functions and DBMS_RLS policies\n05_tests.sql automated PASS/FAIL checks\n06_verification.sql automated PASS/FAIL checks\n07_real_user_smoke_tests.sql actual-user checks\ndemo.sql live demonstration
```

7 Testing and Evaluation

The automated acceptance test checks policy existence, enablement and role-specific row counts. All five policies were enabled for all four statement types. The observed counts were:

Identity	Students	Enrolments	Grades	Payments
Alice Student	1	2	2	1
Bob Student	1	1	1	1
Computing Admin	1	1	2	1
Professor Lee	0	2	2	0
Finance	0	0	0	3
University Admin	3	4	4	3

The output ended with PROJECT I VERIFICATION: PASS. A real-user smoke test additionally confirmed Alice can access one student and two grades, Professor Lee can access two courses and two grades, Finance can access three payments but no grade privilege, and University Admin can access three students and four grades.

8 Visual Verification Result

VPD visibility by application identity

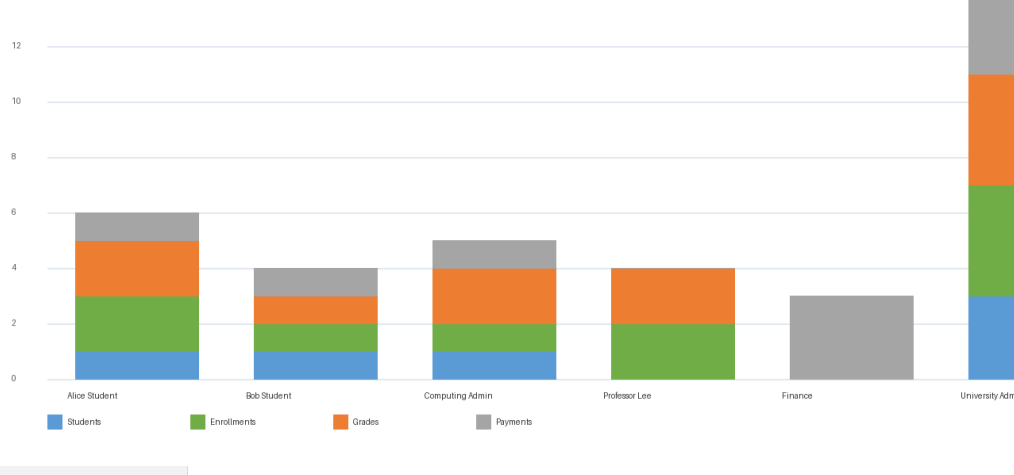


Figure 2. Actual VPD visibility summary used in the acceptance review. Zero values represent intentional denial or out-of-scope data, not failed queries.

9 Demonstration Plan

The live demonstration runs demo.sql. Alice is selected first and only Alice Tan and her two grades are visible. The identity is changed to Bob and the result changes to Bob Lim and one grade. Computing Admin then sees only Computing rows. Finance sees three payments and zero grades. Finally, University Admin sees all three students, four grades and three payments. The team should show both the SQL command and the resulting rows, then explain why each result follows from the policy predicate.

10 Conclusion and Limitations

The project demonstrates that Oracle VPD can enforce personal, course-based, departmental and finance-specific row-level views over shared tables. The combination of ordinary object privileges and VPD predicates provides both coarse- and fine-grained controls.

The dataset is intentionally small and authentication is simulated through the trusted context package for classroom reproducibility. A production system should integrate a secure login service, audit context changes, protect payment data with encryption or tokenisation, and automate regression tests for all write operations.

Contribution Statement

Replace the generic labels below with actual names and keep the percentages truthful. Each member should retain commits, SQL files, test evidence and review records as contribution evidence.

Member	Independent deliverables	Report/demo contribution	Share
Member 1	Application, roles, security requirements, permission matrix	Sections 1-2; requirements explanation	16.7%
Member 2	ER design, tables, constraints, dummy data	Section 3; schema explanation	16.7%
Member 3	Users, roles, grants, context initialization	Section 6; privilege explanation	16.7%
Member 4	VPD functions and DBMS_RLS registration	Section 5; core implementation	16.7%
Member 5	Positive, negative and write-operation tests	Section 7; test evidence	16.7%
Member 6	Runbook, integration, demo script, formatting	Section 9; live demonstration	16.7%

Submission checklist

Confirm the final report is no more than 10 double-column pages excluding the cover and contribution statement. Replace Member 1-6 with real names. Attach or retain the generated verification log and demonstration evidence. Do not submit the local dummy passwords to a shared production environment.